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INDIAN INSTITUTE OF INFORMATION TECHNOLOGY BHOPAL

Department of CSE(Core)/CS/DS/AI/CPS/ECE/P&C/M&C

Branch: - CSE(Core)/CS/DS/AI/CPS/ECE/P&C/M&C **Semester:** - 2nd

End-Term Examination May – 2026

Subject Name: Object Oriented Programming **Max Marks:** 60

Subject Code: CSE/ CSE (CS)/ CSE (DS) /CSE (AI)/ CSE (CPS) /ECE/P&C/M&C-2005 **Duration:** 180 min

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NOTE:

- All questions are compulsory. Write your answers clearly using precise technical words. **Do not leave blank pages between answers, as it may lead to unevaluated responses.** Clearly separate the answers to different questions using sufficient space or lines. **Maintain neat and clean handwriting.**
- While solving, clearly mention the question numbers and their respective parts. Attempt the questions in the given sequence. **All subparts of a question must be answered together in the same place.**
- If any question appears ambiguous, do not worry; clearly state your assumptions in **capital letters** before answering.
- While writing programs, use correct syntax as required in a code editor for successful compilation and execution. Only fully correct programs without syntax errors will be awarded full mark

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1	**A**	What are the names of four pillars of Object-Oriented Programming, explain. Write a single C++ program that demonstrates all four pillars of OOP and clearly identify in your program where and how each of these concepts is implemented.	6
	B	What are constructors and destructors in C++? Explain their uses. Discuss the different types of constructors with suitable examples.	6
2	**A**	What is inheritance in C++? Explain it clearly with an example. List the different types of inheritance and illustrate each with suitable block diagrams. Also, write the general syntax used to define inheritance (all types) in C++, where a derived class inherits from a base class, near the block diagram. Explain how the visibility (accessibility) of base class members changes in a derived class under different types of inheritance in C++	6
	B	**(1)** List the names of any two books related to C++ programming along with their respective authors. **(2)** What is compile time polymorphism? Explain function overloading in detail with the help of suitable examples.	2+4=6

3	A	Write a C++ program to model a payment system. Create a base class **Payment** with a member function **pay() that** displays a general message **"Welcome to our payment system"**. Derive two classes **CreditCard** and **UPI** from Payment. In class **CreditCard**, there is a pay() function to display: "**Payment done using Credit Card**". In class UPI, pay() function exists to display: "**Payment done using UPI**" In the main() function, demonstrates the behavior of the pay() function for objects of all three classes. In addition to this, discuss the OOP concept used in the program clearly. **(Hint: Demonstrate how a single interface can be used to call different implementations at runtime.)**	6
	B	Write a C++ program to model a visitor tracking system for a building. Create a class **VisitorCounter** that maintains the number of visitors currently inside the building using a data member count, initialized to 0. Provide a member function display() to show the current number of visitors. Define a suitable operation such that when a visitor enters the building, the count increases by 1. This operation should be usable in the form **++counter**. Create a derived class **ExitCounter** from **VisitorCounter** that defines a corresponding operation such that when a visitor leaves the building, the count decreases by 1. This operation should be usable in the form --**counter**. In the main() function, create an object named counter of the derived class and perform 3 entry and 2 exit operations using the above forms. Display the number of visitors after the operations. Explain the program and discuss the concepts used in the program in detail. **(Hint: Use inheritance and operator overloading. Implement prefix ++ and -- operators.)**	6
4	A	Write a C++ program using inheritance in which you define a base class **Account** having data members **accountNumber** (integer) and **balance** (float). The value of accountNumber must be assigned only once at the time of object creation and must not be modified later. Derive a class **SavingsAccount** publicly from Account that includes an additional data member **interestRate** (float). All data members must be initialized strictly using constructor initialization lists, and **no assignment using =** should be done inside the constructor body. The base class constructor should initialize accountNumber to 12345 and balance to 10000, while the derived class constructor should initialize interestRate to 5. Implement a member function **display()** in the derived class to print the account number, balance, interest rate, and the calculated interest using the formula **interest = balance * interestRate / 100**. In the main() function, create an object named s1 of class SavingsAccount and call the display() function	6
	B	Define a base class **Person** containing a data member **name** (string). Create two classes **Student** and **Employee** that both inherit from Person and add their own data members **rollNo** (integer) and **empId** (integer) respectively. Further, create a class **TeachingAssistant** that inherits from both Student and Employee and contains an additional data member stipend (float). Initialize the following values using constructorsname = "Amit"; rollNo = 101; empId = 5001; and stipend = 15000,	6

Provide a member function display() in TeachingAssistant to print all details. **Discuss the concept which will be used to run the program efficiently.**

5 A Predict the output of the following program. If the program contains an error, write "Error" and justify your answer. Show necessary steps/calculations. **6**

1

```
#include<iostream>
using namespace std;
class Distance
{
    int meter;
public:
    Distance(int m)
    {
        meter = m;
    }
    Distance(Distance
&obj)
    {
        obj.meter = meter;
    }
    void display() const
    {
        cout << "Meter: "
<< meter << endl;
    }
};
int main()
{
    Distance d1(500);
    Distance d2 = d1;
    d1.display();
    d2.display();
    return 0;
}
```

2

```
#include<iostream>
using namespace std;
class Shape
{
public:
    void draw()
    {
        cout << "Drawing Shape"
<< endl;
    }
};
class Circle : public Shape
{
public:
    void draw()
    {
        cout << "Drawing
Circle" << endl;
    }
};
int main()
{
    Shape *ptr; Circle c;
    Shape s; ptr = &c;
    ptr->draw(); c.draw();
    s.draw();
    Shape &ref1 = c;
    Shape &ref2 = s;
    ref1.draw(); ref2.draw();
    return 0;
}
```

3

```
#include <iostream>
using namespace std;
class A {
public:
    void show(int x) {
  cout << "Function from
class A" << endl;
    }
```

4

```
#include<iostream>
using namespace std;
class p {
    public:
    int i; };
class q: virtual public p {
    public:
    int j; };
```

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```
};
class B {
public:
    void show() {
        cout << "Function
from class B" << endl;
    }
};
class C : public A, public
B { };
int main() {
    C obj;
    obj.show(5);
    return 0;
}
```

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```
class r: virtual public p {
    public:
    int k;};
class s: public p, public q,
public r {
    public:
    int mul;};
int main(){
    s o;
    o.i = 10;
    o.j = 20;
    o.k = 30;
    o.mul = o.i * o.j * o.k;
    cout << o.mul;
    return 0;}
```

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<td>B</td>
<td colspan="2">For the following programs, some parts of the code are missing. Complete only the missing statements as per the instructions given in each program. Do not rewrite the complete program, only write the missing code properly and clearly.</td>
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<tr>
<td>1</td>
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```
#include <iostream>
using namespace std;
class Base {
    int pvt;
public:
    Base(int a) {
        pvt = a;
    }
    int getPVT() {
        return pvt;
    }
};
class PrivateDerived :
private Base {
public:
    _____Here, write
missing statement or
statements needed to
complete and run the code
successfully_______

    int access_helper()
    {
        return getPVT();
    }
};
int main() {
```

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```
#include<iostream>
using namespace std;
class Time{
    int minutes;
public:
    Time(int m){
        minutes = m;
    }
//(1) Write conversion code
______________________________
};
int main(){
    Time t1(90);
    // (2) Implicit  type
conversion
    int x = ____________;
    // (3) Explicit type
conversion
    int y =____________;
    cout << x << " " << y;
    return 0;
}
```

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	`` PrivateDerived object1(10); cout<< object1.access_helper(); return 0; } ```	
3	see below	4

3

```
#include<iostream>
using namespace std;
class Demo {
public:
    int x;
    Demo(int a){
        x = a;
    }
    void show(int y){
        cout << "Sum: " <<
x + y << endl;
    }
};
int main(){
    Demo d(10);
/* Below, write statement
or statements needed to
complete and run the code
successfully */

// Pointer ptr1 to data
member

_____________________;

// Pointer ptr2 to member
function

_____________________;

// Access data member

cout << _________ << endl;

// Call member function
using pointer
_____________________
return 0;
}
```

4

```
#include<iostream>
using namespace std;

class Distance {
    int meter;
public:
    Distance(int m) {
        meter = m;
    }

    // Declare friend operator+
for two Distance objects
_____________________________;

    void display() {
        cout << meter << endl;
    }
};

/* Write the complete
definition of operator+
function */
_____________________________ {
    return Distance(
_____________________________ );
}

int main() {
    Distance d1(5), d2(10);

    // Use overloaded operator
to add two objects
    Distance d3 =
_____________________________;

    d3.display();
    return 0;
}
```

				// Hint: Operator overloading using friend function. This program adds two distances.